MICHAEL SHAHIN

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EDUCATION

University of Kansas, Lawrence, KS *2019 - 2024*
PhD *with Honors*, Geology (Advisors: Leigh Stearns & C.J. van der Veen)
Dissertation: Where the Ice Ends: Calving and Ice-Ocean Interactions at Helheim Glacier and Beyond

College of Charleston, Charleston, SC *2014 - 2018*
BS *magna cum laude*, Geology

PROFESSIONAL APPOINTMENTS

Postdoctoral Researcher *2025 - Present*
University of Pennsylvania, Philadelphia, PA

Research Affiliate *2025 - Present*
Center for Remote Sensing and Integrated Systems (CReSIS), Lawrence, KS

PUBLICATIONS

Submitted and In Revision

Rines, J.H., C.Y., Lai, E. Abrahams, **M.G. Shahin**, N.B., Coffey, E. Lee, L.A, Stevens, L.A, Stearns. A benchmark deep learning dataset for the classification of supraglacial lake drainage mechanism across the central-west Greenland Ice Sheet. *Submitted to EGU Earth System Science Data*. https://doi.org/10.5194/essd-2026-406

Shahin, M.G., K. Hughes, L. A. Stearns, C.J. van der Veen, A. Roth, F. Straneo. Can a glacier protect itself? Viewing sikkusak (ice mélange) and deep-keeled icebergs as a heat sink. *Under Review at AGU Advances*. https://essopenarchive.org/doi/full/10.22541/essoar.15004066/v1

Published and Accepted

Shionalyn, K., G. Catania, D. Trugman, **M.G. Shahin**, D. Felikson, L.A. Stearns. Outlet Glacier Seasonal Terminus Prediction Using Interpretable Machine Learning. 2026. *The Cryosphere*. https://tc.copernicus.org/articles/20/1725/2026/

Shahin M.G., L.A. Stearns, C.J. van der Veen, D.C. Finnegan, A.L. LeWinter, S.F. Child, S. O'Neel, H. Butler. Calving Mechanisms Inferred from Observations of Surface Depressions at Helheim Glacier, Greenland. 2025. *Journal of Geophysical Research: Earth Surface*, 130, e2024JF008059. https://doi.org/10.1029/2024JF008059

Harcourt, W. D., **M.G. Shahin**, L.A. Stearns, and S. Shankar: Structural weaknesses in ice mélange revealed by high resolution ICEYE SAR imagery. 2025. *Journal of Glaciology*. https://doi.org/10.1017/jog.2025.10085

Meng, Y., C.Y., Lai, R. Culberg, **M.G. Shahin**, L.A. Stearns, J. Burton, K. Nissanka. Seasonal Changes of Mélange Thickness Coincide With Greenland Calving Dynamics. 2025. *Nature Communications*. https://doi.org/10.1038/s41467-024-55241-7

Fahrner, D., G. Catania, **M.G. Shahin**, D.D. Hansen, K. Löffler, J. Abermann. Advances in monitoring glaciological processes in Kalaallit Nunaat (Greenland) over the past decades. 2024. *PLOS Climate*. https://doi.org/10.1371/journal.pclm.0000379

Melton, S. M., R.B. Alley, S. Anandakrishnan, B.R. Parizek, **M.G. Shahin**, L.A Stearns, A. L. LeWinter, D. C. Finnegan. 2022. Meltwater drainage and iceberg calving observed in high-spatiotemporal resolution at Helheim Glacier, Greenland. *Journal of Glaciology*. https://doi.org/10.1017/jog.2021.141

DATASETS

- Autonomous Terrestrial Laser Scanner Point Clouds, Velocities, and Elevation Models
  - Accepted as an Amazon Web Services Public Dataset
  - `aws s3 ls --no-sign-request s3://atlas-lidar-helheim/`
- Helheim Glacier Grounding Lines and Grounding Zones
  - https://doi.org/10.5281/zenodo.15368065

ORAL PRESENTATIONS

Shahin, M. G., K. Hughes, L. A. Stearns, A. Roth, F. Straneo. Can a glacier protect itself? Viewing sikkusak (ice mélange) as a heat sink. KU Physics & Astronomy Locally Organized Assembly 2024.

Shahin, M. G.. Where the Ice Ends: Calving and Ice-Ocean Interactions at Helheim Glacier and Beyond. University of Texas Institute for Geophysics Discussion Hour 2024. *Invited*.

Shahin, M. G., L. A. Stearns. Helheim Glacier Heising-Simons Research Update. Future of Greenland Ice Sheet Science 2023.

Shahin, M. G., L. A. Stearns, D. C. Finnegan, A. L. LeWinter, H. Butler, C.J. van der Veen, S. O'Neel. How Does Glacier Geometry Dictate Calving Style? An Analysis of Helheim Glacier Using Two Autonomous Terrestrial Laser Scanners. In AGU Fall Meeting 2021. AGU.

Shahin, M. G. Calving initiation and response at Helheim Glacier derived from repeat autonomous Terrestrial LiDAR Scanners. Midwest Glaciology Meeting 2021.

Shahin, M. G., L. A. Stearns, D. C. Finnegan, A. L. LeWinter, C.J. van der Veen, H. Butler. Insights Into Helheim Glacier Terminus Dynamics Through Integration of Two Autonomous Terrestrial LiDAR Scanners, 2015-2019. In AGU Fall Meeting 2020. AGU.

CONFERENCE ABSTRACTS

Shahin M.G., L.A. Stearns, C.J. van der Veen, D.C. Finnegan, A.L. LeWinter, H. Butler. Glacier and ice mélange velocity products using a new point cloud registration algorithm from two autonomous terrestrial laser scanners. In AGU Fall Meeting 2024. AGU.

Child, S.F., L.A. Stearns, **M.G. Shahin**, C.J. van der Veen. Historic ice thickness changes of Helheim Glacier using 1940s aerial photography. In AGU Fall Meeting 2024. AGU.

Shahin, M. G., S. Shankar, L. A. Stearns, K. Hughes, C.J. van der Veen, F. Straneo. Mélange as a heat sink: Mapping sikkusak (ice mélange) ice-ocean heat transfer around the Greenland Ice Sheet. In AGU Fall Meeting 2023. AGU.

Shionalyn, K., G. Catania, D. Trugman, D. Felikson, L.A. Stearns, **M.G. Shahin**. Predicting and Analyzing Greenland Outlet Glacier Seasonality Using Machine Learning. In AGU Fall Meeting 2023. AGU.

Meng, Y., C.Y., Lai, R. Culberg, **M.G. Shahin**, L.A. Stearns, J. Burton, K. Nissanka. Thickness of Pro-glacial Mélange Impacts Calving Dynamics of Greenland Glaciers. In AGU Fall Meeting 2023. AGU.

Harcourt, W. D., L.A. Stearns, **M.G. Shahin**, and S. Shankar: Assessment of ice mélange impacts on tidewater glacier dynamics using high resolution ICEYE imagery, EGU General Assembly 2023, Vienna, Austria, 24–28 Apr 2023, EGU23-9812, https://doi.org/10.5194/egusphere-egu23-9812, 2023.

Meng, Y., R. Culberg, **M.G. Shahin**, L.A. Stearns, J. Burton, K. Nissanka, C.Y., Lai. Thickness of Pro-glacial Mélange Impacts Calving Dynamics of Greenland Glaciers. Bulletin of the American Physical Society, 2023.

Mejia, J.Z, L.A. Stearns, J. Bassis, L.M. Simkins, M.M Lummus, C.T. Barnett, **M.G. Shahin**, J.W. Burton, S.A. Goliber, R. Duddu, E. Ultee, C. Trunz, N. Stevens. CryoCommunity: On the need for strategic goals to guide EDI efforts across the polar sciences. In AGU Fall Meeting 2022. AGU.

Shahin, M. G., S. Shankar, L. A. Stearns, D. C. Finnegan, A. L. LeWinter, H. Butler, C.J. van der Veen, S. O'Neel. Does Iceberg Configuration Impact the Behavior of Helheim Glacier's Ice Mélange? In AGU Fall Meeting 2022. AGU.

Harcourt, W.D., J. Hiral, A. Wehrlé, A. Tarvi, A. Kuye, A, Tolipov, D. Hallé, H. Guímaro, L. Taylor, **M.G. Shahin**, M. Saha, N, Pereira, S. Jawak, S. Shankar, S. Duveau, V. Goel, W.J. Leong, B. O'Halloran. The APECS Polar Earth Observation Database. In AGU Fall Meeting 2022. AGU.

Shahin, M. G., S. Shankar, L. A. Stearns, D. C. Finnegan, A. L. LeWinter, H. Butler, C.J. van der Veen, S. O'Neel. Calving at Helheim Glacier dominated by large flexure events: Observations from two autonomous terrestrial laser scanners. IGS International Symposium on Maritime Glaciers 2022.

Mejia, J.Z, C.T. Barnett, J. Bassis, R. Duddu, S. Goliber, M.M. Lummus, **M.G. Shahin**, L.M. Simkins, L.A. Stearns, E. Ultee. Best practices for building a more inclusive glaciology through cryocommunity.org. In AGU Fall Meeting 2021. AGU.

Stearns, L.A., S. Shankar, **M.G. Shahin**, C.J. van der Veen. Distribution and motion of icebergs in a proglacial melange: weekly observations from 2020 at Helheim Glacier, East Greenland. In AGU Fall Meeting 2021. AGU.

Shahin, M. G., L. A. Stearns, D. C. Finnegan, A. L. LeWinter, A. S. Fowler. When does iceberg calving initiate? Insights into flexure zone formation at Helheim Glacier using two autonomous terrestrial LiDAR scanners. KU GHawk Symposium. 2021.

Melton, S. M., R.B. Alley, S. Anandakrishnan, B.R. Parizek, **M.G. Shahin**, L.A Stearns, A. L. LeWinter, D. C. Finnegan. Iceberg Calving and Meltwater Drainage Observed in High Spatial and Temporal Resolution at Helheim Glacier, Greenland. In AGU Fall Meeting 2021. AGU.

Harcourt, W. D., L.A. Stearns, **M.G. Shahin**, and S. Shankar. Mapping icebergs in melange using ICEYE imagery. In AGU Fall Meeting 2021. AGU.

Stearns, L.A., S. Shankar, **M.G. Shahin**, S. Rezvanbehbahani, C.J. van der Veen. Mélange variability around Greenland yields insights into calving dynamics and conditions that favor mélange formation. In AGU Fall Meeting 2020. AGU.

Shahin, M. G., L. A. Stearns, D. C. Finnegan, A. L. LeWinter, A. S. Fowler, P. J. Gadomski, & C. J. van der Veen. Investigating the Relationship Between Iceberg Calving and Terminus Velocity at Helheim Glacier from Two Autonomous Terrestrial LiDAR Scanners. In AGU Fall Meeting 2019. AGU.

Shahin, M. G., L. A. Stearns, D. C. Finnegan, A. L. LeWinter, A. S. Fowler. Glacier Velocity Response to Iceberg Calving at an East Greenland Glacier from Terrestrial LiDAR Scanners. KU GHawk Symposium. 2019.

Shahin, M. G., L. A. Stearns, D. C. Finnegan, A. L. LeWinter, P. J. Gadomski, A. S. Fowler. 2019. Observations of Ice-Ocean Interactions at Helheim Glacier from Two Terrestrial LiDAR Scanners. 27th IUGG General Assembly. Montreal, Canada.

ACADEMIC SERVICE & OUTREACH

- Reviewer
  - Geophysical Research Letters, The Cryosphere
- Helheim Glacier Art Installation *2025*
- International Glaciological Society Early Career Researcher Committee Member *2025 -*
- Geology Graduate Student Organization Vice President *2023 - 2024*
- C13B AGU Fall Meeting 2023 Session Chair *2023*
- International Arctic Science Committee Podcast Episode *2023*
- MARS Unlearning Racism in the Geosciences Pod and CryoCommunity *2021 - 2022*
- AWG Outreach Coordinator *2019 - 2022*
  - Develop outreach modules for the Association for Women Geoscientists
- University of Kansas Department of Geology DEI Committee Member *2021*
- What do glaciers have to do with coding?! *2020*
  - An interview with content creator and iOS developer Mayuko Inoue

HONORS & AWARDS

- Erasmus Haworth Graduate Honors, $2000, *KU Department of Geology* *2025*
- KU Geology Summer Research Scholarship, $6000, *KU Department of Geology* *2024*
- GHawk Symposium 1st Place, $700, *KU Department of Geology* *2023*

- Frederick T. Holden Scholarship, $4,500 *KU Department of Geology* *2023*
- D.A McGee Scholarship, $3,500 *KU Department of Geology* *2022*
- Best Student Presentation Runner-Up *International Glaciological Society* *2022*
- FitzSimmons Scholarship, $1,250, *KU Department of Geology* *2019*
- GHawk Symposium 1st Place, $700, *KU Department of Geology* *2019*
- CofC Geology Departmental Honors, *CofC* *2018*
- CofC Geology Outstanding Student Award, *CofC* *2018*
- SC Space Grant Undergraduate Fellowship, $12,000, *CofC* *2016-2018*
- GeoCUR Award for Excellence in Student Research, *CofC* *2017*
- HHMI Research Mentor Award, $500, *CofC* *2017*
- WEASC Legacy of Learning Scholarship, $1,750, *CofC* *2017*
- NASA EPSCoR Palmetto Research Academy, $5,000, *CofC* *2016*

CERTIFICATES

- Geographic Information Systems Graduate Certificate *2019 - 2021*
  - University of Kansas, Lawrence, KS

GLACIOLOGY FIELD EXPERIENCE AND SUMMER SCHOOLS

- Machine learning in glaciology summer school (GlaMacLeS) *June 2024*
- Hokkaido University cryosphere modeling summer school *May–June 2024*
- Helheim Glacier, Southeast Greenland *Sept. 2022*
  - Installation of time-lapse cameras and terrestrial laser scanner maintenance
- Greenland Ice Sheet Ocean Science Network Summer School *July 2022*
- McCarthy International Summer School in Glaciology
  - Accepted in 2020, canceled due to COVID
- UNIS AG 325: Glaciology *Feb. - March 2020*
  - Glaciology field course in Svalbard
- Helheim Glacier, Southeast Greenland *May 2019*
  - Assisted in the maintenance of two autonomous terrestrial laser scanner systems

COMPUTATIONAL SKILLS

- **Languages:** `Python`, `SQL/PostgreSQL/PostGIS`, `Bash`
- **Scientific Python:** `xarray`, `NumPy`, `Numba`, `SciPy`, `pandas/GeoPandas`, `Matplotlib`
- **Scalable / out-of-core:** `Dask`, `Zarr`, `COPC`, `obstore`, `Polars`
- **Cloud & infrastructure:** AWS (`S3`, `EC2`, Open Data Registry), Linux/Unix, `git`/GitHub
- **Geospatial & remote sensing:** `QGIS`, `STAC`, `GDAL`, `PDAL`, ice-penetrating radar, terrestrial LiDAR, SAR, ICESat-2, PostGIS
- **Statistical & ML methods:** Bayesian changepoint detection, signal filtering, Monte Carlo
- **Markup Languages:** LaTeX, Markdown